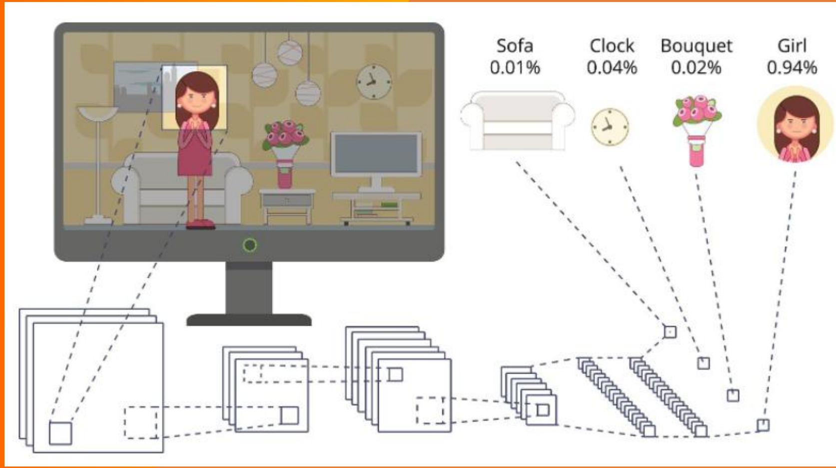
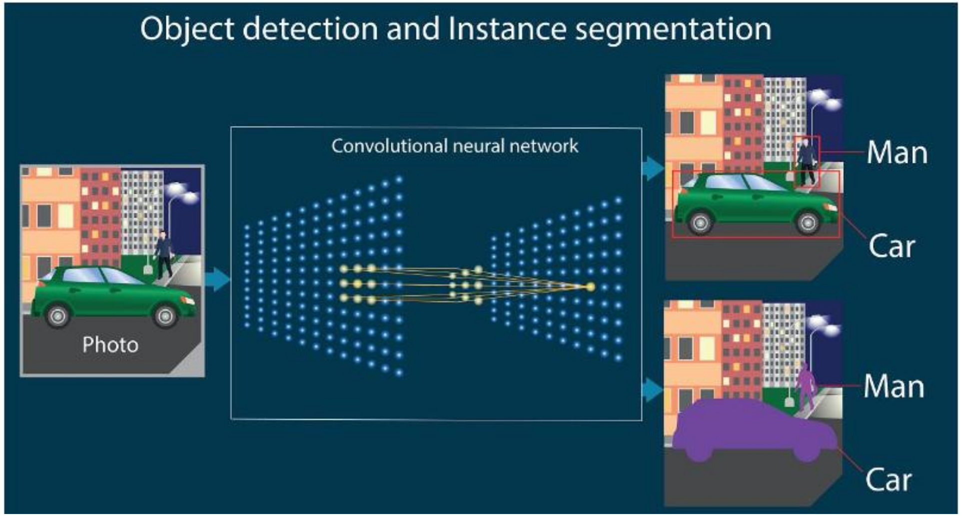
Module 05: Convolutional Neural Network: How can a computer recognise a cat when it has never actually seen one?



Courtesy to Andrew Ng Deep Learning Course DeepLearning.AI and https://cs.uwaterloo.ca/~mli/Deep-Learning-2017-Lecture5CNN.ppt

Learning Outcomes/Agenda

- Week 4 recap
- Why ordinary neural networks struggle with images
- How humans recognise objects
- Understanding CNNs
- Convolution and Filters
- Feature Maps
- Pooling
- Flattening and Classification
- CNN Architectures
- Interactive Python Demonstration
- Reflection and Discussion



Module-04 Recap

- Why Models Fail? (Generalisation)
- Underfitting vs Overfitting
- Regularization Intuition
- L1 vs L2 Regularization
- Interactive Python Demonstration
- GenAI Critique Activity
- Reflection & Key Takeaways

Recap Quiz

Question 1

- **What is the main purpose of regularisation in machine learning?**
- A. Increase the number of hidden layers
- B. Reduce overfitting and improve generalisation
- C. Increase the training error
- D. Make the network larger

Answer: B

Discussion: Why is a model that performs well on new (unseen) data more valuable than one that simply memorises the training data?

Question 2

- **Which statement best describes an overfitted model?**
- A. It performs very well on the training data but poorly on unseen data.
- B. It performs equally well on all datasets.
- C. It cannot learn any meaningful patterns.
- D. It always has high training and testing errors.

Answer: A

Discussion: Can you think of a real-life situation where memorisation is less useful than understanding?

Recap

Question 3

- **Which regularisation method encourages some weights to become exactly zero, effectively selecting only the most important features?**
- A. Gradient Descent
- B. Batch Normalisation
- C. L2 Regularisation
- D. L1 Regularisation

Answer: D

Discussion: Why might removing unnecessary features improve a model's ability to generalise?

Question 4

- **What is the primary difference between L1 and L2 regularisation?**
- A. L1 increases the learning rate, while L2 decreases it.
- B. L1 sets some weights to zero, while L2 shrinks weights without making them exactly zero.
- C. L2 removes neurons, while L1 adds neurons.
- D. There is no difference.

Answer: B

- **Discussion**
- In what situations might you prefer L1 over L2?

Transition

Question 5 (Transition to CNNs)

- **Images often contain millions of pixels. Based on what you learned about model complexity and overfitting, why might using a fully connected neural network directly on large images be problematic?**
- A. Images are too colourful.
- B. It would require a very large number of parameters, increasing computation and the risk of overfitting.
- C. Images cannot be processed by neural networks.
- D. Images only contain grayscale information.

Answer: B

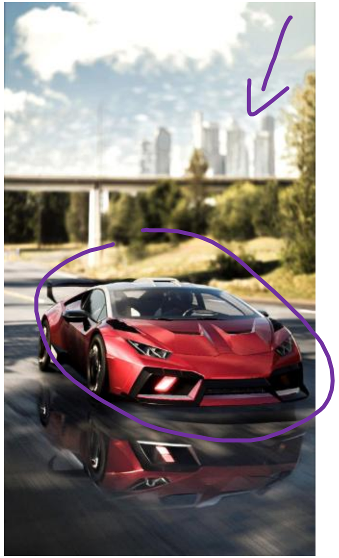
Transition

- Last week we learned that large models with many parameters can easily overfit. Today we'll explore how Convolutional Neural Networks (CNNs) solve this challenge by learning image features much more efficiently, using fewer parameters while achieving higher accuracy.

Assessment Progression

- Assessment 2 is due at the end of Module 8 and Assessment 3 is due at the end of Module 12. Please refer to the assessment briefs in the assessment area for more detail.
- Tasks to undertake in this Module to prepare for Assessments 2 and 3 include:
  - Identifying possible uses of CNNs for your projects;
  - Analysing specific CNN algorithms for your projects; and
  - Examining Python implementation of
- You can prepare for these assessments by using the learning resources of this Module, which include readings, videos and learning activities. These resources will enhance your understanding of the key concepts in this Module.

Can Computers Really "See"?



What features help YOU recognise a car?
How might a computer recognise these same objects?

How do we teach a computer to extract meaningful visual features from raw pixels?

- Human Vision

Image

↓

Eyes

↓

Brain

↓

Edges

Shapes

Patterns

↓

Object

- Computer Vision

Image

↓

Pixels

(Thousands of Numbers)

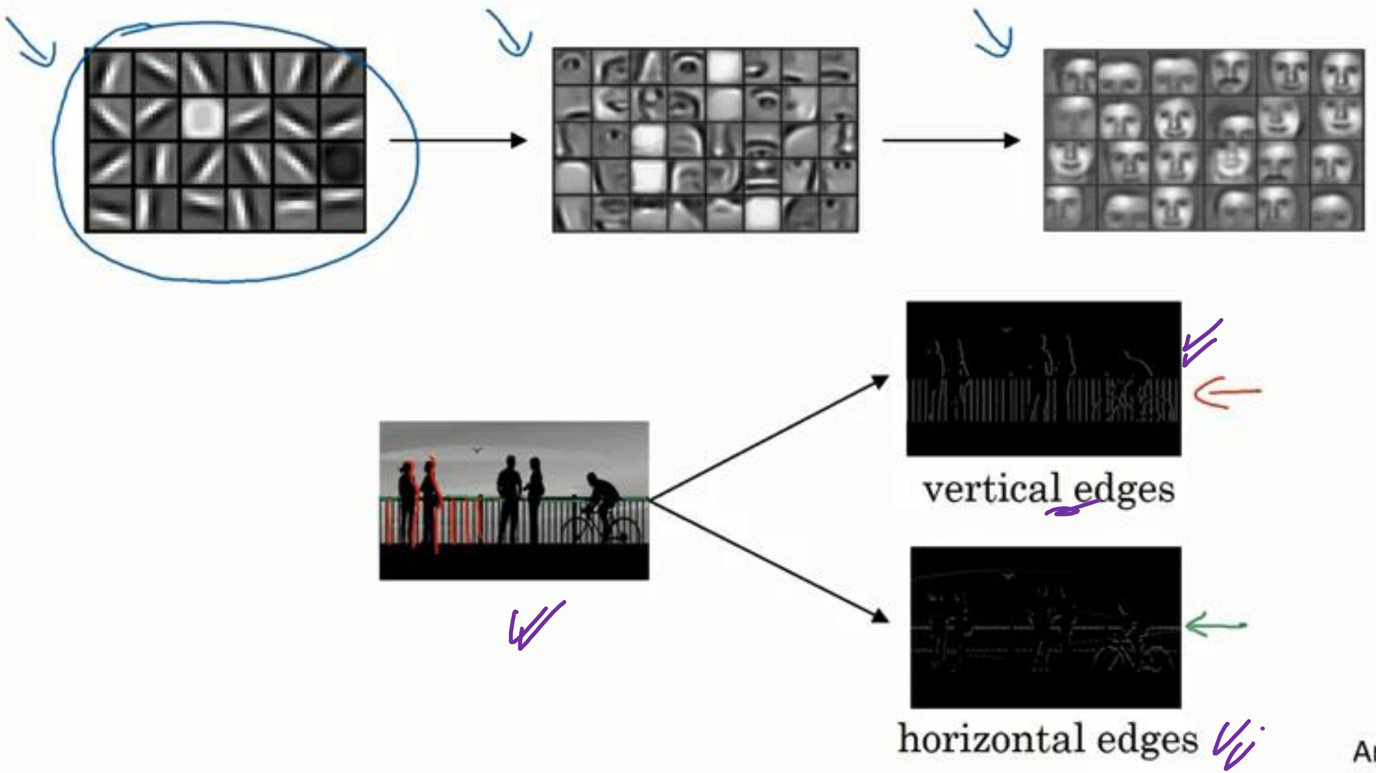
↓

?????

↓

Prediction

Computer Vision Problem



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Vertical edge detection

$3\times1 + 1\times1 + 2\times1 + 0\times0 + 5\times0 + 7\times0 + 1\times-1 + 8\times-1 + 2\times-1 = -5$

"convolution"

$$\underbrace{\begin{bmatrix} 3 & 0 & 1 & 2 & 7 & 4 \\ 1 & 5 & 8 & 9 & 3 & 1 \\ 2 & 7 & 2 & 5 & 1 & 3 \\ 0 & 1 & 3 & 1 & 7 & 8 \\ 4 & 2 & 1 & 6 & 2 & 8 \\ 2 & 4 & 5 & 2 & 3 & 9 \end{bmatrix}}_{\boxed{6\times6}} * \underbrace{\begin{bmatrix} 1 & 0 & -1 \\ 1 & 0 & -1 \\ 1 & 0 & -1 \end{bmatrix}}_{\substack{3\times3 \\ \text{filter}}} = \underbrace{\begin{bmatrix} -5 & -4 & 0 & 8 \\ -10 & -2 & 2 & 3 \\ 0 & -2 & -4 & -7 \\ -3 & -2 & -3 & -16 \end{bmatrix}}_{4\times4}$$

python: conv-forward

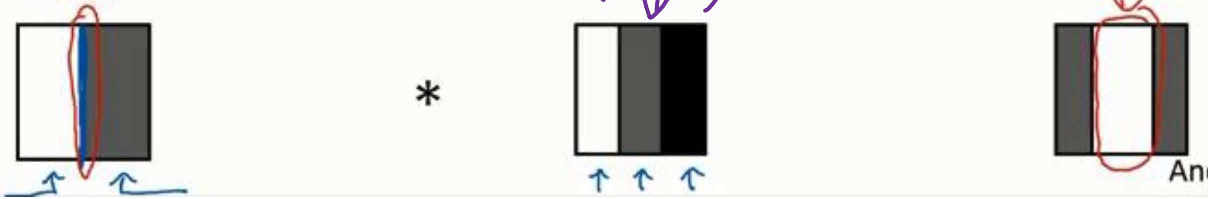
tensorflow: tf.nn.conv2d

keras: Conv2D

Andrew Ng

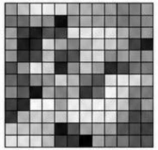
Vertical Edge Detection

$$\underbrace{\begin{bmatrix} 10 & 10 & 10 & 0 & 0 & 0 \\ 10 & 10 & 10 & 0 & 0 & 0 \\ 10 & 10 & 10 & 0 & 0 & 0 \\ 10 & 10 & 10 & 0 & 0 & 0 \\ 10 & 10 & 10 & 0 & 0 & 0 \\ 10 & 10 & 10 & 0 & 0 & 0 \end{bmatrix}}_{6 \times 6} * \underbrace{\begin{bmatrix} 1 & 0 & -1 \\ 1 & 0 & -1 \\ 1 & 0 & -1 \end{bmatrix}}_{3 \times 3} = \underbrace{\begin{bmatrix} 0 & 30 & 30 & 0 \\ 0 & 30 & 30 & 0 \\ 0 & 30 & 30 & 0 \\ 0 & 30 & 30 & 0 \end{bmatrix}}_{4 \times 4}$$

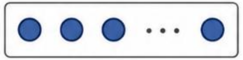


Why Feedforward Networks Struggle

Using a Traditional Neural Network on Images

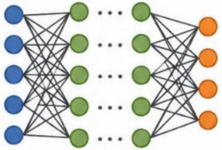


64 × 64 Image



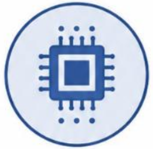
4096 Input Neurons

(64 × 64 = 4096 pixels)



Thousands of Hidden Connections

Fully connected layers lead to an explosion of connections



Millions of Parameters

High computational cost, more data required, higher risk of overfitting

Why are Convolutional Neural Networks (CNNs) Important?

Traditional AI struggled with images because...

Millions of pixels Every image contains thousands or millions of pixel values that must be processed.

Huge number of parameters Fully connected neural networks require an enormous number of weights, making training computationally expensive.

Poor scalability As image size increases, computation and memory requirements grow rapidly, increasing the risk of overfitting.

CNNs made possible...

Face Recognition Unlock smartphones and identify people in photos.

Self-driving Cars Detect roads, pedestrians, vehicles, and traffic signs in real time.

Medical Diagnosis Detect tumours, fractures, and other abnormalities in medical images.

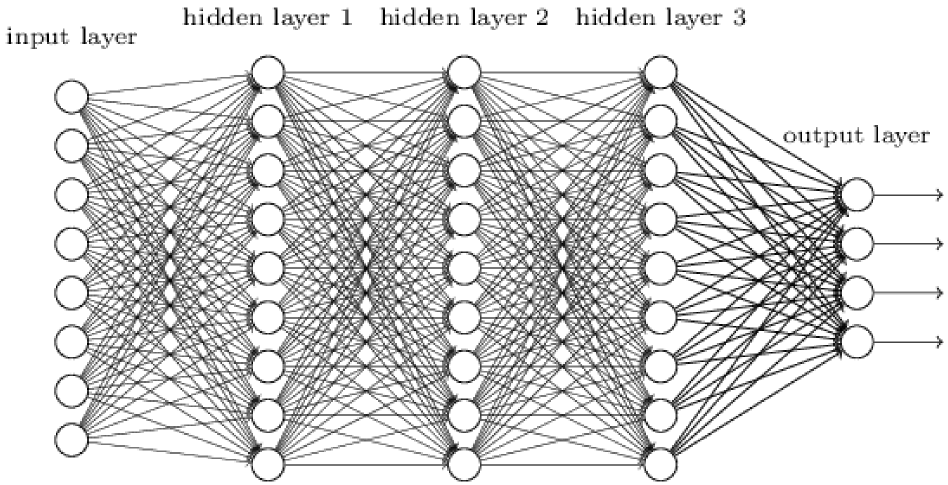
Satellite Monitoring Monitor crops, forests, disasters, and environmental change.

Industrial Inspection Automatically identify defects in manufactured products.

CNNs transformed Computer Vision in much the same way that Large Language Models (LLMs) transformed Natural Language Processing (NLP).

Smaller Network: CNN

- We know it is good to learn a small model.
- From this fully connected model, do we really need all the edges?
- Can some of these be shared?



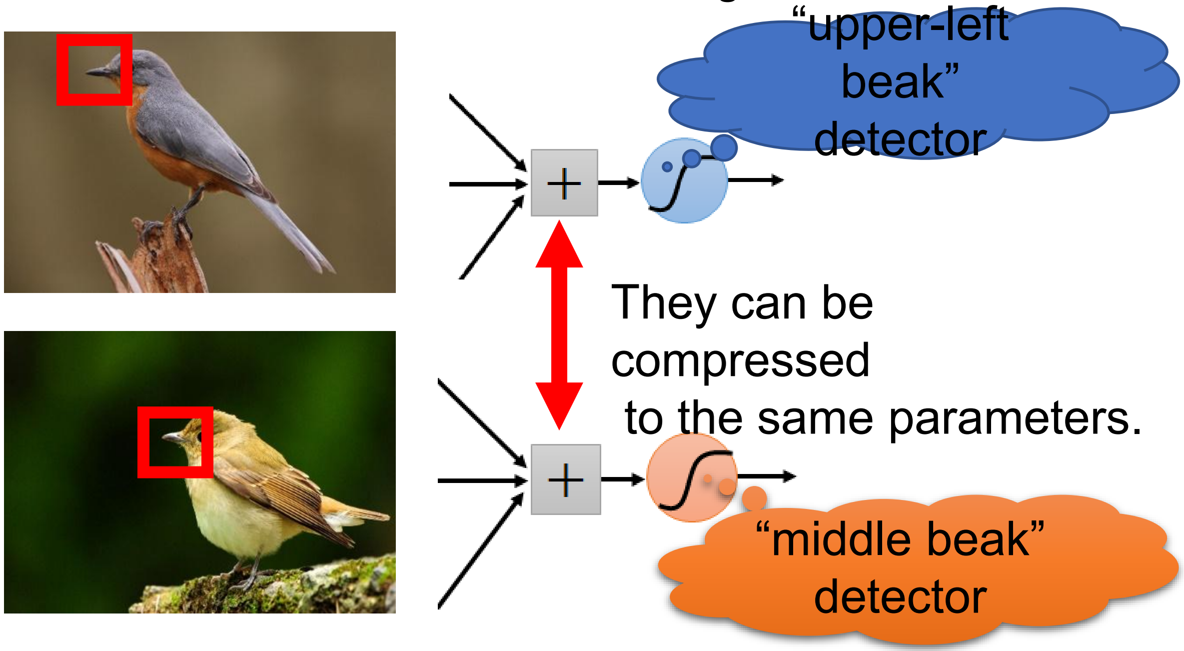
Consider learning an image:

- Some patterns are much smaller than the whole image

Can represent a small region with fewer parameters

Same pattern appears in different places:
They can be compressed!
What about training a lot of such "small" detectors
and each detector must "move around".

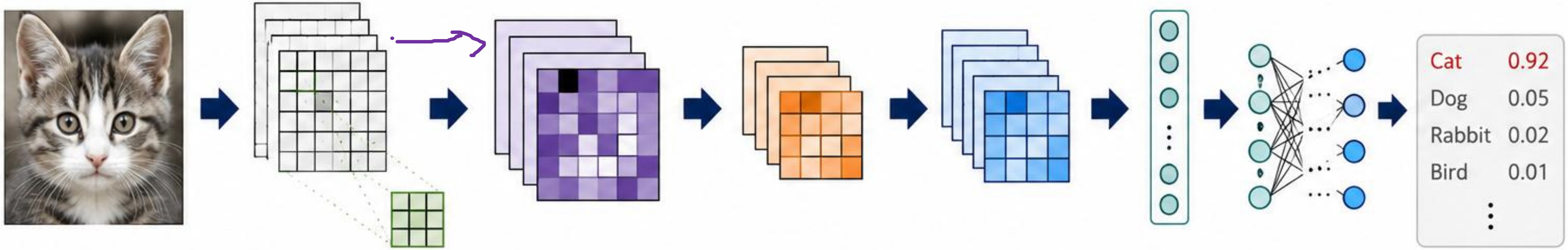
Instead of learning thousands of different beak detectors,
CNNs learn **one detector** and slide it across the entire image.



CNN PIPELINE

From Pixels to Prediction

A CNN transforms an image through a sequence of operations to learn *hierarchical features* and make a prediction.



1. INPUT IMAGE

Raw pixels (e.g., 64 × 64 × 3)

2. CONVOLUTION (Filters)

Filters (kernels) slide over the image to detect local patterns (edges, textures, etc.)

3. FEATURE MAPS (+ ReLU)

Produces feature maps that highlight where patterns are patterns. ReLU adds non-linearity.

4. POOLING (Downsampling)

Reduces the spatial size (e.g., max pooling) to keep imported. information and reduce computation.

5. CONV + POOL (Repeat)

Multiple convolutional and pooling layers learn increasingly complex features.

6. FLATTEN

Converts the final feature maps into a 1D vector.

7. FULLY CONNECTED (DENSE)

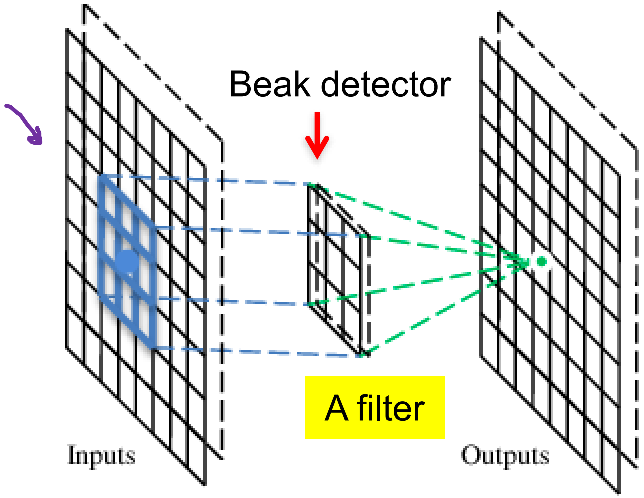
The vector is passed through fully connected layers to learn combinations of features.

8. PREDICTION (OUTPUT)

The output layer produces the final prediction (e.g., class probabilities).

A convolutional layer

A CNN is a neural network with some convolutional layers (and some other layers). A convolutional layer has a number of filters that does convolutional operation.



Convolution

These are the network parameters to be learned.

1	0	0	0	0	1
0	1	0	0	1	0
0	0	1	1	0	0
1	0	0	0	1	0
0	1	0	0	1	0
0	0	1	0	1	0

6 x 6 image

1	-1	-1
-1	1	-1
-1	-1	1

Filter 1

-1	1	-1
-1	1	-1
-1	1	-1

Filter 2

⋮ ⋮

Each filter detects a small pattern (3 x 3).

Convolution

1	-1	-1
-1	1	-1
-1	-1	1

Filter 1

stride=1

1	0	0	0	0	1
0	1	0	0	1	0
0	0	1	1	0	0
1	0	0	0	1	0
0	1	0	0	1	0
0	0	1	0	1	0

6 x 6 image

Dot product → 3 -1

$$\{\underbrace{n-f+1}_{6 \to 4}, \underbrace{n-f+1}\}$$

Convolution

1	-1	-1
-1	1	-1
-1	-1	1

Filter 1

If stride=2

1	0	0	0	0	1
0	1	0	0	1	0
0	0	1	1	0	0
1	0	0	0	1	0
0	1	0	0	1	0
0	0	1	0	1	0

6 x 6 image

3 -3

Convolution

1	-1	-1
-1	1	-1
-1	-1	1

Filter 1

stride=1

1	0	0	0	0	1
0	1	0	0	1	0
0	0	1	1	0	0
1	0	0	0	1	0
0	1	0	0	1	0
0	0	1	0	1	0

6 x 6 image

3	-1	-3	-1
-3	1	0	-3
-3	-3	0	1
3	-2	-2	-1

$6-3+1$

$(4,4)$

Convolution

-1	1	-1
-1	1	-1
-1	1	-1

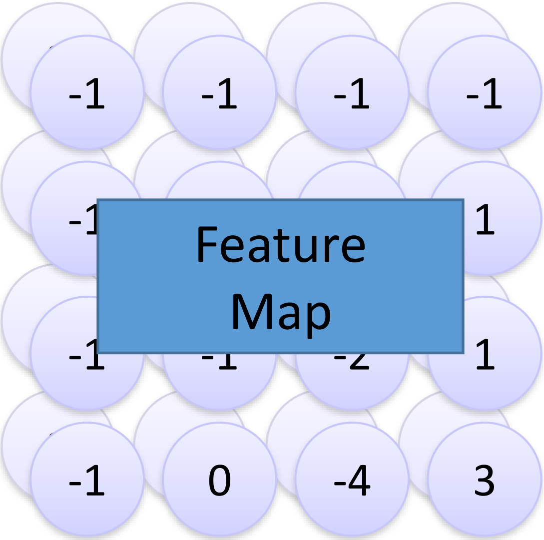
Filter 2

Repeat this for each filter

stride=1

1	0	0	0	0	1
0	1	0	0	1	0
0	0	1	1	0	0
1	0	0	0	1	0
0	1	0	0	1	0
0	0	1	0	1	0

6 x 6 image



Two 4 x 4 images
Forming 2 x 4 x 4 matrix

Color image: RGB 3 channels

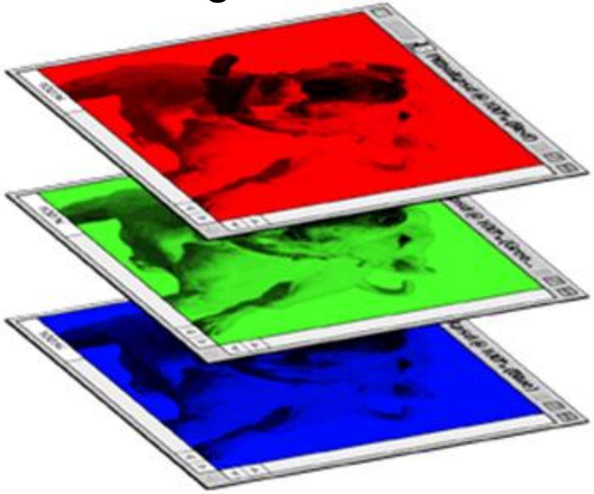
1	-1	-1
-1	1	-1
-1	-1	1

Filter 1

-1	1	-1
-1	1	-1
-1	1	-1

Filter 2

Color image



1	0	0	0	0	1
0	1	0	0	1	0
0	0	1	1	0	0
1	0	0	0	1	0
0	1	0	0	1	0
0	0	1	0	1	0

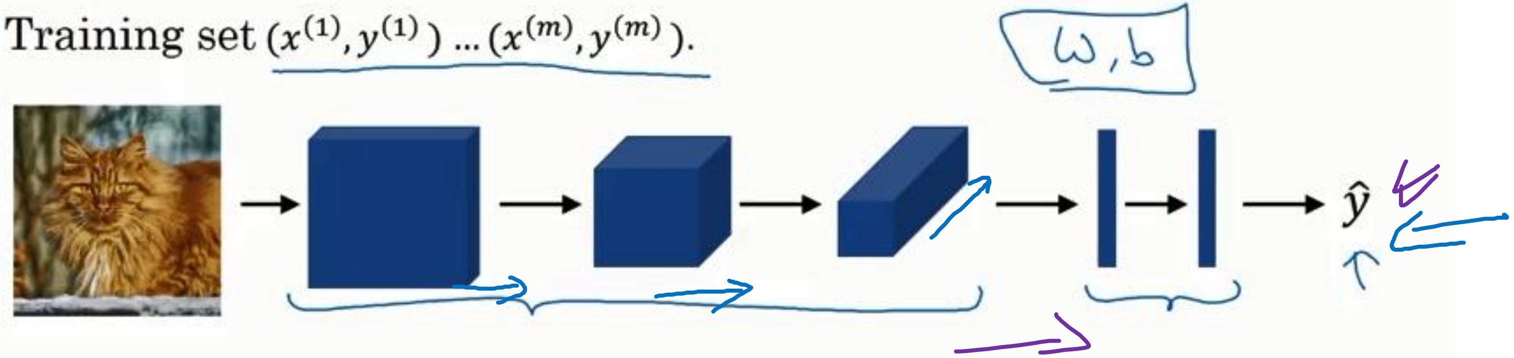
Stop & Think

- Who design the filters?
- Why don't we use one huge filter instead of many small filters?

Putting it together

Putting it together

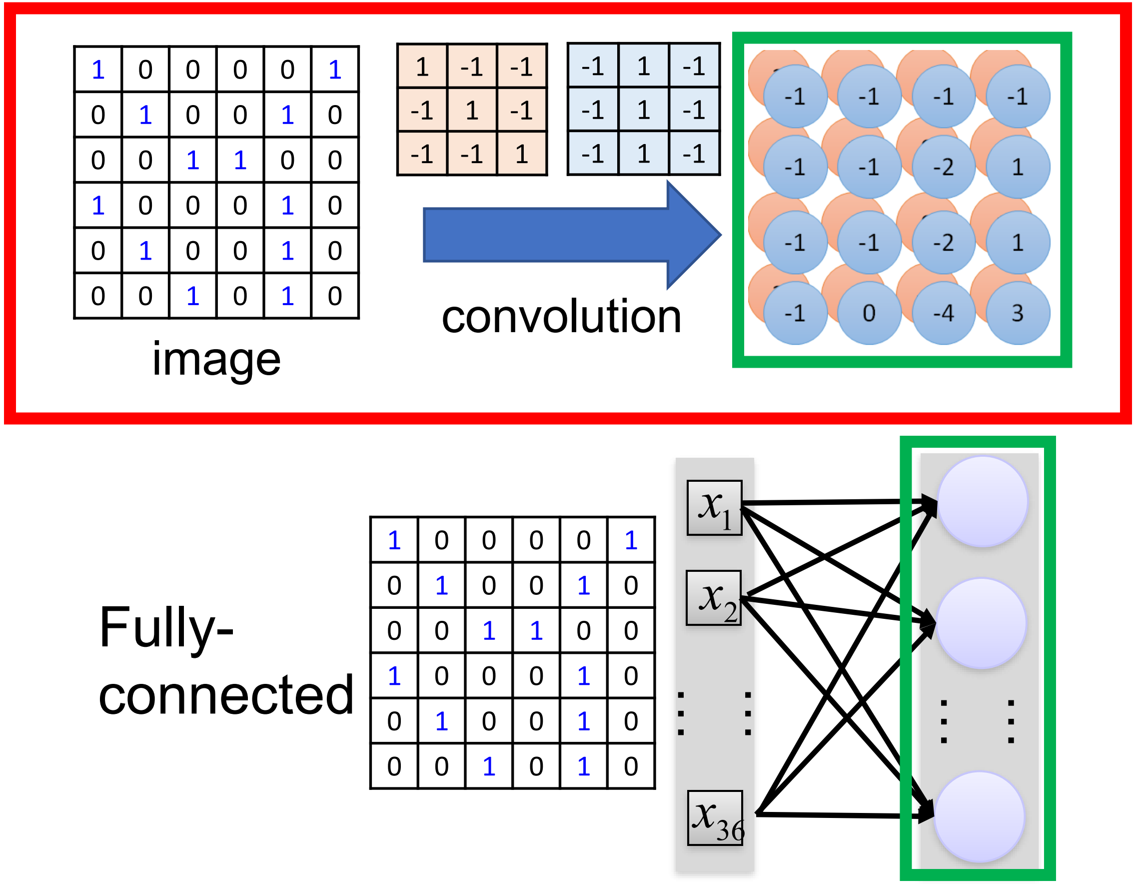
Training set $(x^{(1)}, y^{(1)}) \dots (x^{(m)}, y^{(m)})$.

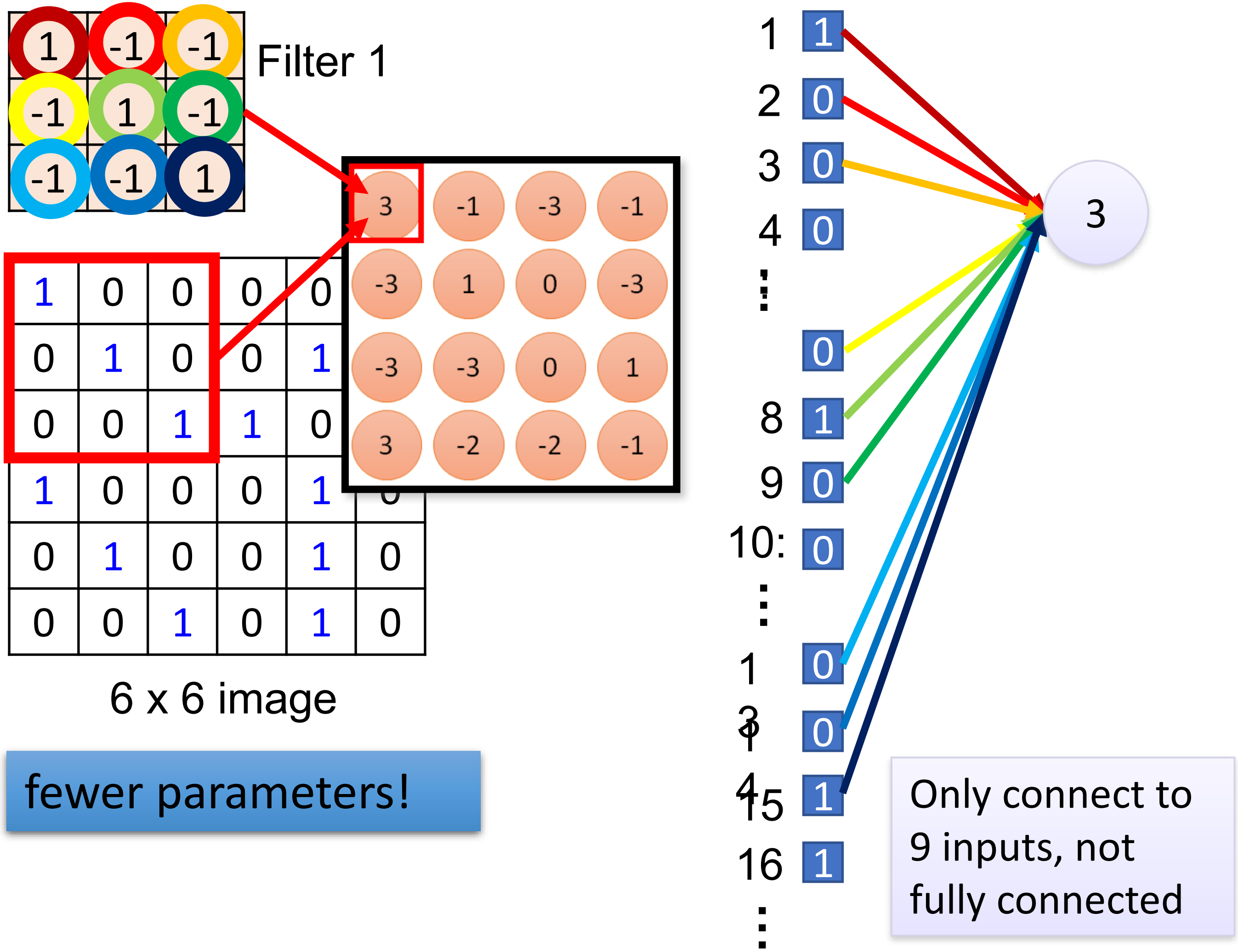


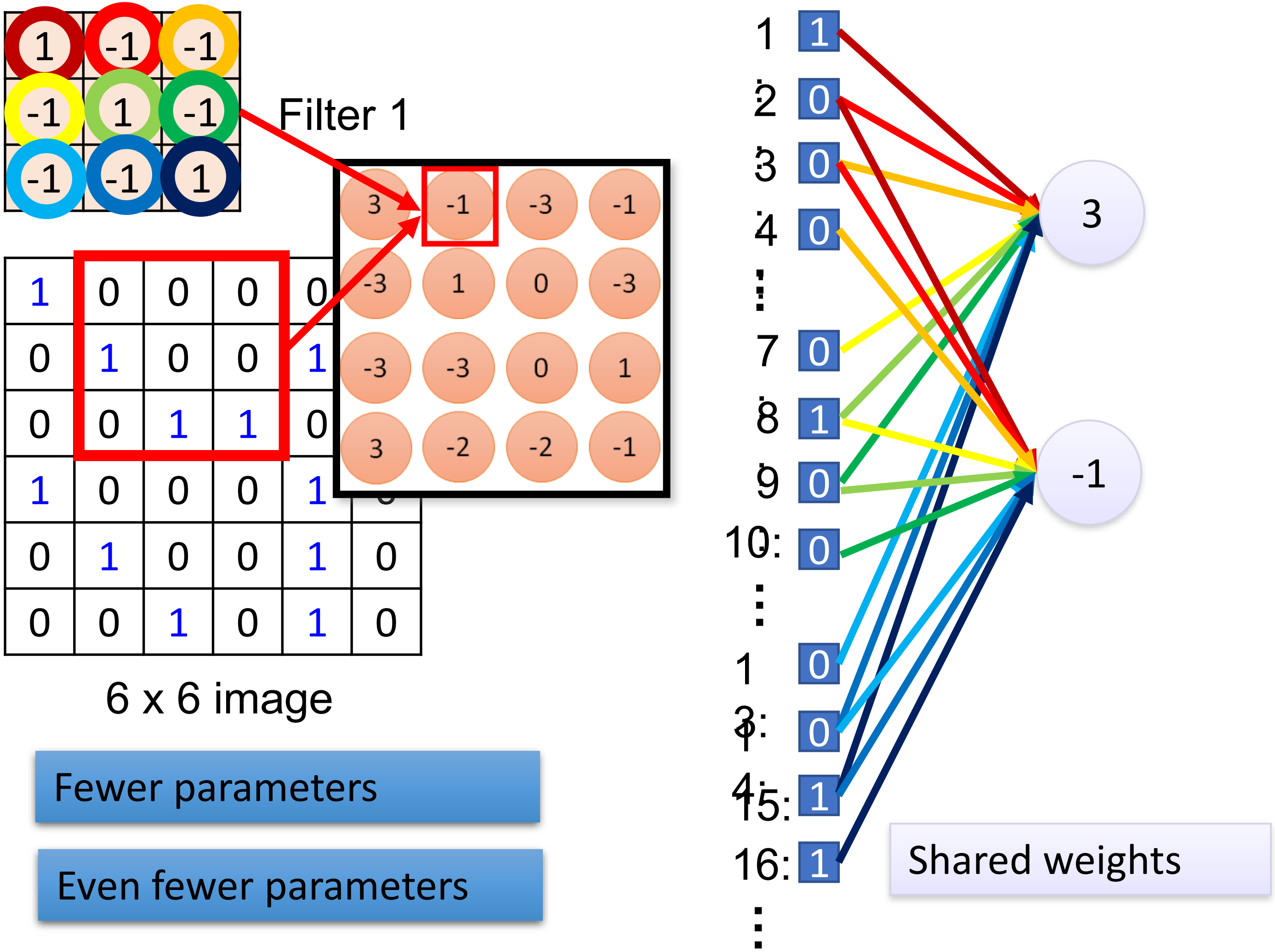
Cost $J = \frac{1}{m}\sum_{i=1}^{m} \mathcal{L}(\hat{y}^{(i)}, y^{(i)})$

Use gradient descent to optimize parameters to reduce J

Convolution v.s. Fully Connected

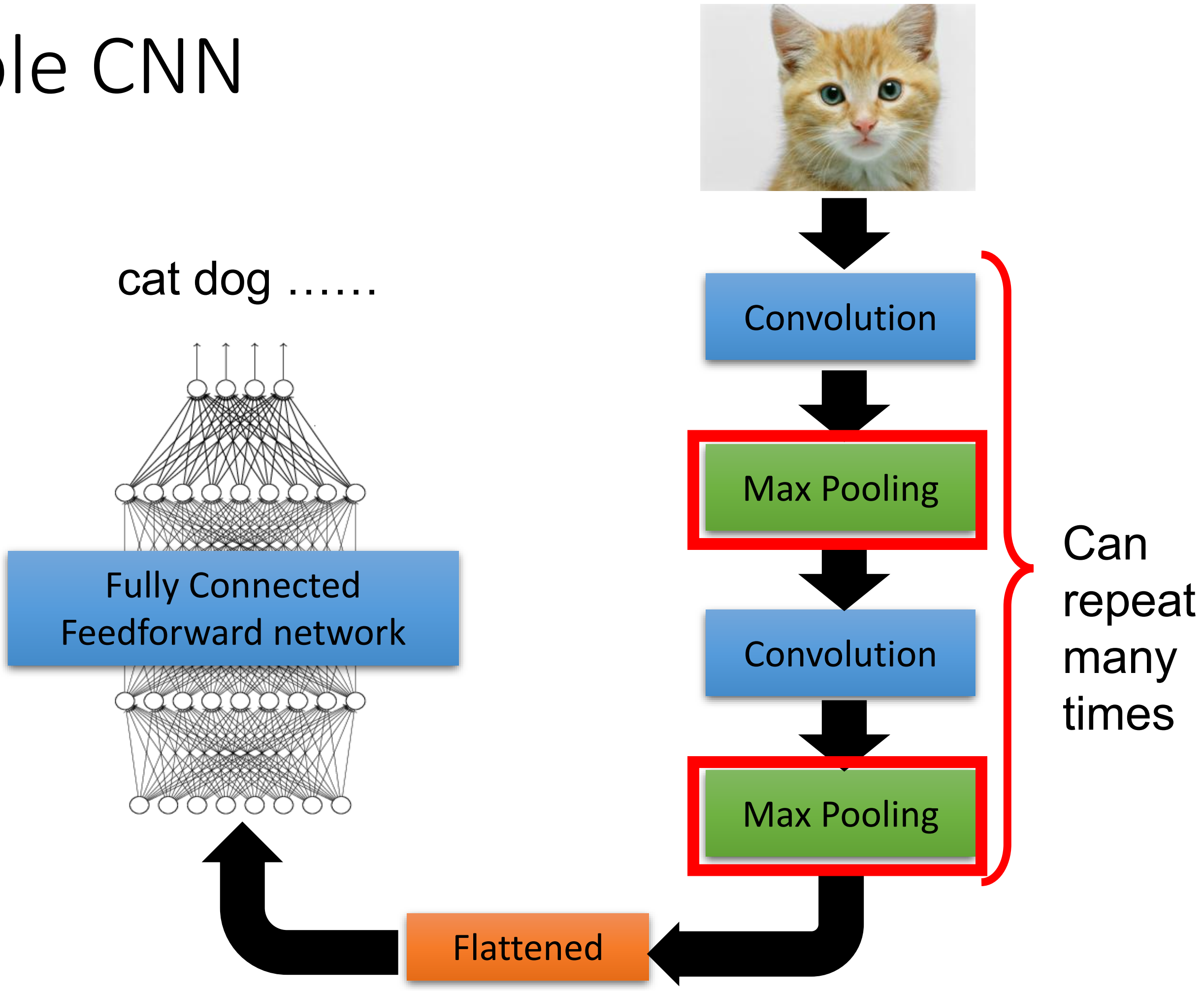



Filter 1
6 x 6 image
fewer parameters!
3
Only connect to
9 inputs, not
fully connected
1
2
3
4
8
9
10:
13
14
15
16



Filter 1
6 x 6 image
Fewer parameters
Even fewer parameters
Shared weights

The whole CNN



Max Pooling

1	-1	-1
-1	1	-1
-1	-1	1

Filter 1

-1	1	-1
-1	1	-1
-1	1	-1

Filter 2

3	-1	-3	-1
-3	1	0	-3
-3	-3	0	1
3	-2	-2	-1

-1	-1	-1	-1
-1	-1	-2	1
-1	-1	-2	1
-1	0	-4	3

Why Pooling

What information do you think is lost during pooling?

- Subsampling pixels will not change the object

bird

Subsampling

bird

We can subsample the pixels to make image smaller i.e., fewer parameters to characterize the image

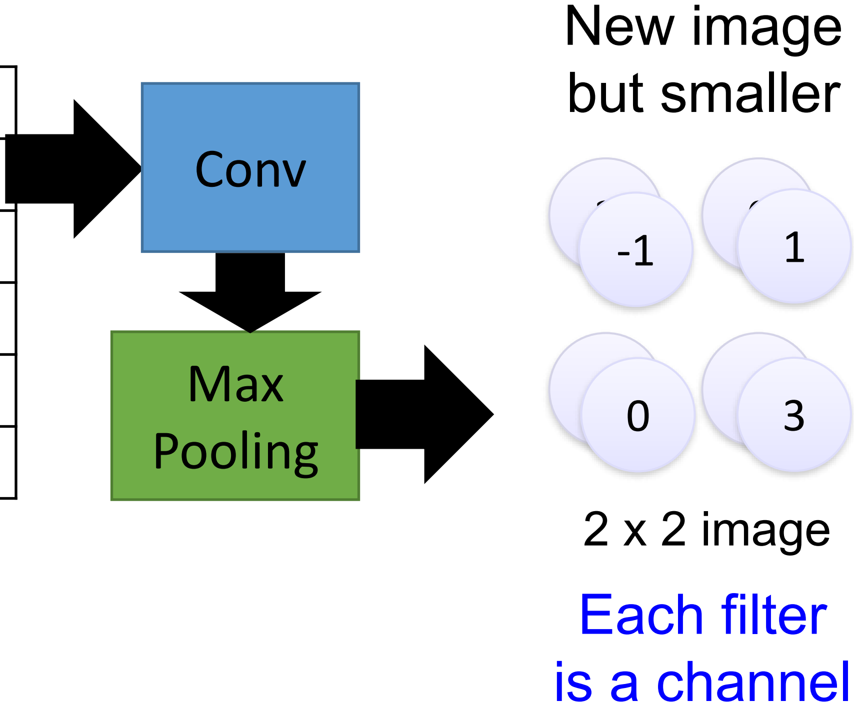
A CNN compresses a fully connected network in two ways:

- Reducing number of connections
- Shared weights on the edges
- Max pooling further reduces the complexity

Max Pooling

1	0	0	0	0	1
0	1	0	0	1	0
0	0	1	1	0	0
1	0	0	0	1	0
0	1	0	0	1	0
0	0	1	0	1	0

6 x 6 image



What has the CNN learned so far?

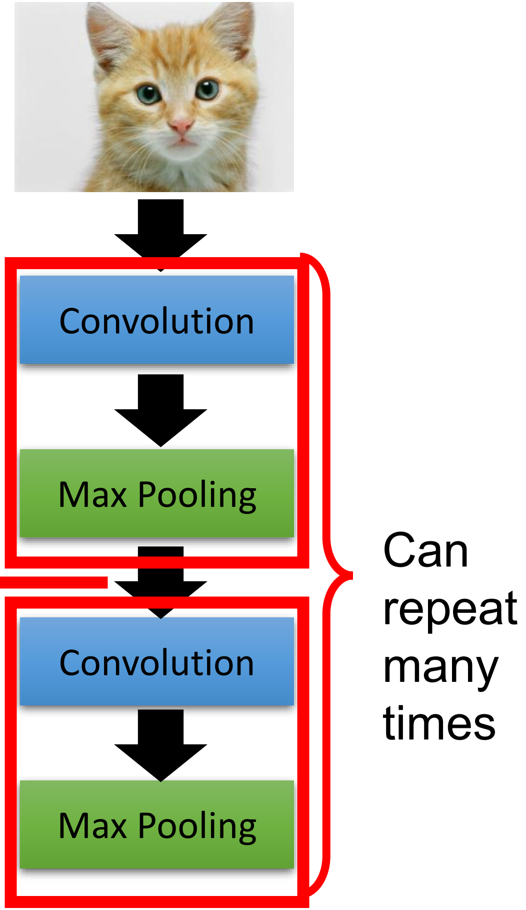
CNNs learn **hierarchical visual features.**

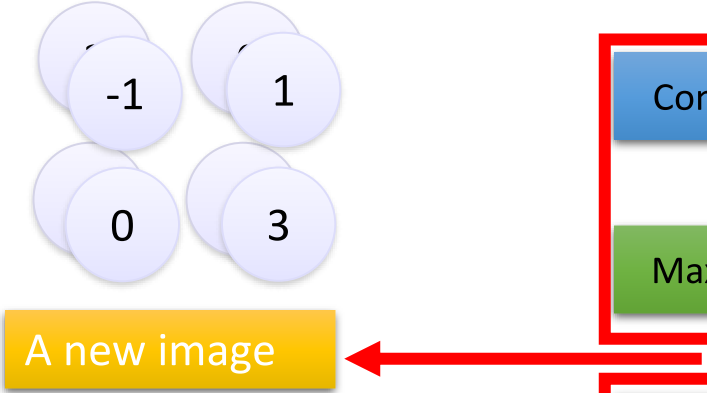
Each deeper layer builds on simpler features to recognise more complex patterns and objects.

Image	Edge detector	Corner detector	Texture detector	Eye detector	Face detector	Cat detector

Image	Edge detector	Corner detector	Texture detector	Eye detector	Face detector	Cat detector
Raw pixels (colours and intensities)	Detects simple edges and orientations	Detects corners and junctions	Detects textures and repetitive patterns	Detects parts like eyes, nose, ears, etc.	Detects whole faces by combining parts	Recognises the cat as a complete object

The whole CNN

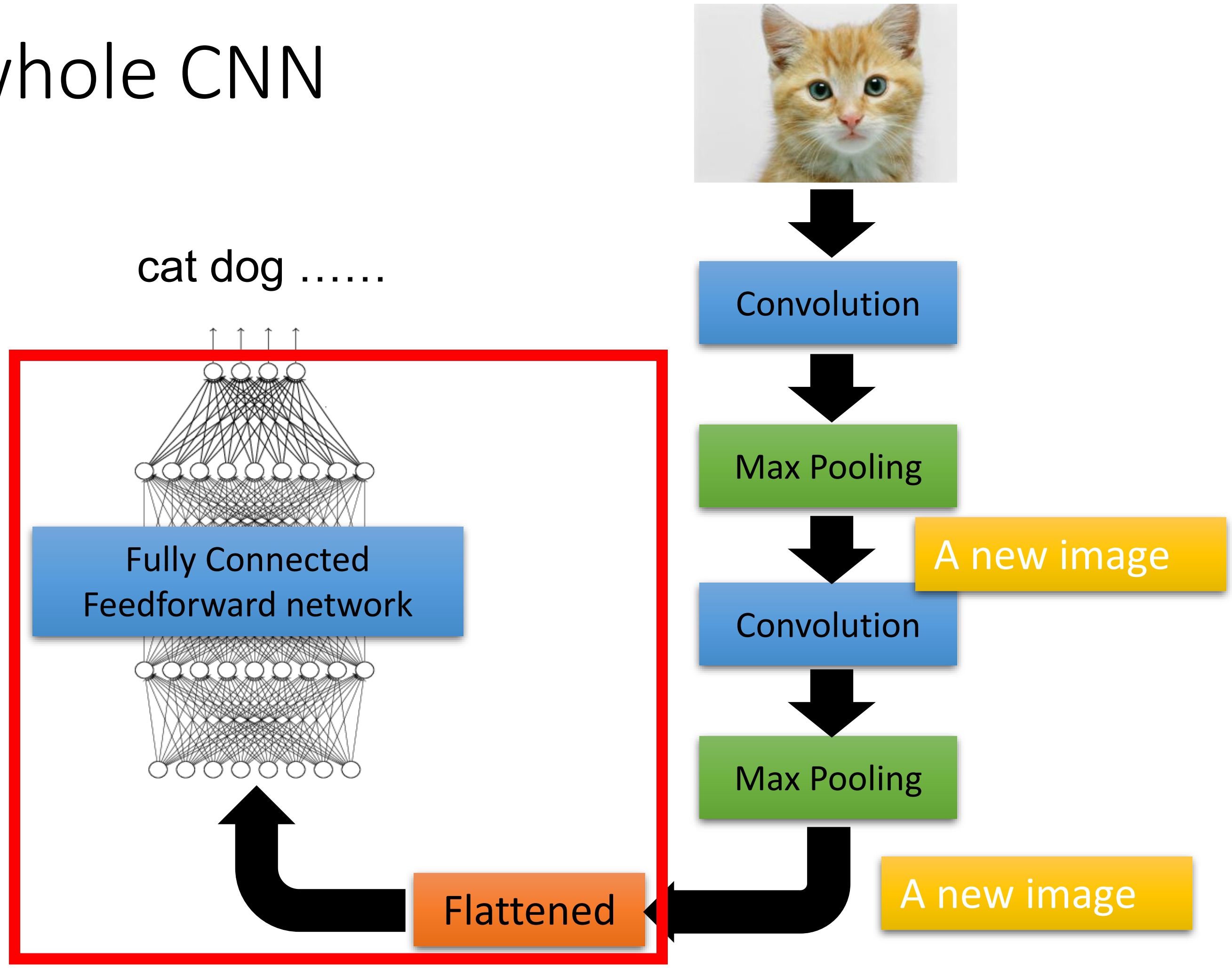




Smaller than the original image

The number of channels is the number of filters

The whole CNN



AlphaGo: Why do you think DeepMind chose CNNs instead of a traditional neural network?

19 x 19 matrix

Black: 1

white: -1

none: 0

→ Neural Network →

Next move

(19 x 19 positions)

Fully-connected feedforward network can be used

But CNN performs much better

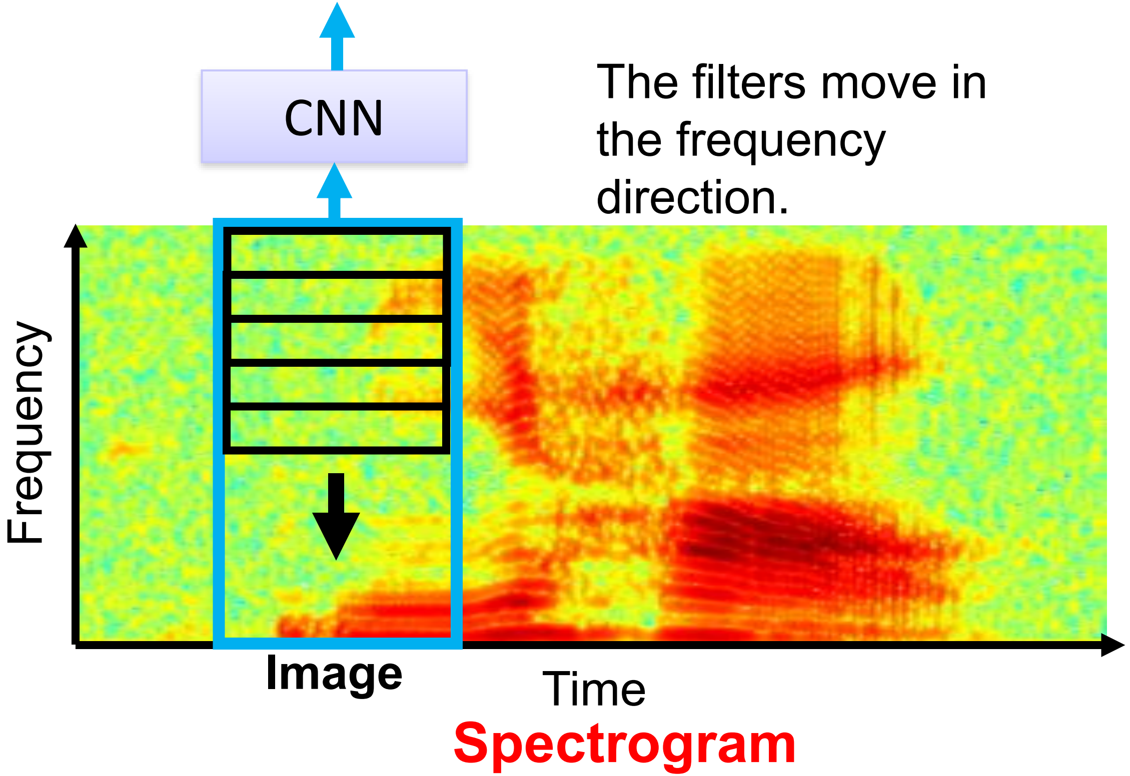
AlphaGo's policy network

The following is quotation from their Nature article:

Note: AlphaGo does not use Max Pooling.

Neural network architecture. The input to the policy network is a $19 \times 19 \times 48$ image stack consisting of 48 feature planes. The first hidden layer zero pads the input into a 23×23 image, then convolves k filters of kernel size 5×5 with stride 1 with the input image and applies a rectifier nonlinearity. Each of the subsequent hidden layers 2 to 12 zero pads the respective previous hidden layer into a 21×21 image, then convolves k filters of kernel size 3×3 with stride 1, again followed by a rectifier nonlinearity. The final layer convolves 1 filter of kernel size 1×1 with stride 1, with a different bias for each position, and applies a softmax function. The match version of AlphaGo used $k = 192$ filters; Fig. 2b and Extended Data Table 3 additionally show the results of training with $k = 128$, 256 and 384 filters.

CNN in speech recognition



The filters move in the frequency direction.

Python Activity: Experiment Challenge

Try changing:

- the filter
- the stride
- the pooling size

- Which change had the biggest effect?
- Which filter best detected edges?
- Which filter blurred the image?
- Which configuration would you use for medical images?

Learning Activity: Discussion Forum Post

- Using the deep learning resources, you have read in this Module, please identify at least two projects or research ideas that you would like to pursue as part of this unit.
- Describe each of these in one clear problem statement and write associated challenges or research questions for each statement. In this activity, you should demonstrate your ability to identify a research gap, propose a project to address the gap and communicate details in short written form. It should also help you to connect with likeminded people in the class.
- **Post your ideas to the Module 1 discussion forum.**
- **Please read other students' posts and provide feedback on their ideas. Most importantly, identify other students who are planning to work on projects similar to yours.**